

Scalable Federated and Parallel Algorithms for Unique Column Combination Discovery

This research investigates whether it is possible to efficiently discover minimal hitting sets for Unique Column Combination (UCC) in large datasets within federated environments and distributed systems. The core objective is to design a scalable architecture that integrates federated processing and parallel computation (Client X Core) to accelerate enumeration algorithms for data profiling.

The study is based on the hitting set enumeration algorithm proposed by Bläsius et al., which is widely used for discovering minimal UCCs in relational datasets. In database systems, UCC discovery is important for identifying candidate keys and supporting schema analysis, data quality assessment, and database optimization. The algorithm explores the search space of candidate column combinations through a decision-tree enumeration process with pruning via an extension oracle.

To improve the computational efficiency of this generation and enumeration process, the research explores two complementary strategies. For **federated environment and distributed computation** will use **ALMA many-core cluster at the University of Vienna**.

First, the data will be distributed in multiple systems for computation model toward a **federated environment**. Instead of centralizing the dataset, the data is partitioned across multiple simulated nodes. Each node performs the generation process, where data will be converted into dummy coding to prevent sharing real data and transmitted to the host server coordinator, where aggregation is performed to derive global minimal hitting sets.

Second, parallel execution is used to accelerate the enumeration process. The original algorithm, implemented in C++ and sequential, will be converted into distributed and parallel computing and will be executed on the **ALMA many-core cluster at the University of Vienna**, where multiple CPU cores process different branches of the enumeration tree simultaneously. Parallel task distribution significantly reduces runtime for complex datasets by exploiting multi-core computation to perform aggregation to derive global minimal hitting sets, and it will mainly focus on time and task management.

The overall framework integrates both approaches: local parallel computation and distributed federated aggregation. By combining these techniques, the system aims to reduce computational complexity while maintaining the correctness of the resulting UCC discovery process.

The final goal of the research is to efficiently compute **minimal hitting sets from datasets that do not contain explicit primary keys**, enabling automatic discovery of candidate keys in large or distributed data environments. The approach is designed to operate iteratively on smaller partitions or groups of data, allowing scalable analysis of large datasets.